

API to a newer compatibility API. An example of the same can be seen below:

```
test-pc:~$cat test-infer-v1.py

# Tensorflow imports
import tensorflow as tf

save_path = 'checkpoints/dev'
with tf.gfile.FastGFile("./trained-graph.pb", 'rb') as f:
    graph_def = tf.GraphDef()
    graph_def.ParseFromString(f.read())
    tf.import_graph_def(graph_def, name='')

    with tf.Session(graph=tf.get_default_graph()) as sess:
        input_data = sess.graph.get_tensor_by_
name("DecodeJPGInput:0")
        output_data = sess.graph.get_tensor_by_name("final_
result:0")

        image = 'elephant-299.jpg'
        if not tf.gfile.Exists(image):
            tf.logging.fatal('File does not exist %s', image)
        image_data = tf.gfile.FastGFile(image, 'rb').read()

        result = sess.run(output_data, {'DecodeJPGInput:0':
image_data})
        print(result)

test-pc:~$ tf_upgrade_v2 --infile test-infer-v1.py --outfile
test-infer-v2.py

INFO line 5:5: Renamed 'tf.gfile.FastGFile' to 'tf.compat.
v1.gfile.FastGFile'
INFO line 6:16: Renamed 'tf.GraphDef' to 'tf.compat.
v1.GraphDef'
INFO line 10:9: Renamed 'tf.Session' to 'tf.compat.
v1.Session'
INFO line 10:26: Renamed 'tf.get_default_graph' to 'tf.
compat.v1.get_default_graph'
INFO line 15:15: Renamed 'tf.gfile.Exists' to 'tf.io.gfile.
exists'
INFO line 16:12: Renamed 'tf.logging.fatal' to 'tf.compat.
v1.logging.fatal'
INFO line 17:21: Renamed 'tf.gfile.FastGFile' to 'tf.compat.
v1.gfile.FastGFile'
TensorFlow 2.0 Upgrade Script
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Converted 1 files
Detected 0 issues that require attention
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Make sure to read the detailed log 'report.txt'
```

```
test-pc:~$ cat test-infer-v2.py

# Tensorflow imports
import tensorflow as tf

save_path = 'checkpoints/dev'
with tf.compat.v1.gfile.FastGFile("./trained-graph.pb", 'rb')
as f:
    graph_def = tf.compat.v1.GraphDef()
    graph_def.ParseFromString(f.read())
    tf.import_graph_def(graph_def, name='')

    with tf.compat.v1.Session(graph=tf.compat.v1.get_default_
graph()) as sess:
        input_data = sess.graph.get_tensor_by_
name("DecodeJPGInput:0")
        output_data = sess.graph.get_tensor_by_name("final_
result:0")

        image = 'elephant-299.jpg'
        if not tf.io.gfile.exists(image):
            tf.compat.v1.logging.fatal('File does not exist
%s', image)
        image_data = tf.compat.v1.gfile.FastGFile(image,
'rb').read()

        result = sess.run(output_data, {'DecodeJPGInput:0':
image_data})
        print(result)
```

As we can see here, the *tf_upgrade_v2* utility converts all the deprecated APIs to compatible v1 APIs, to make them work with 2.0.

Eager execution: Eager execution allows real-time evaluation of Tensors without calling *session.run*. A major advantage with eager execution is that we can print the Tensor values any time for debugging.

With TensorFlow 1.x, the code is:

```
test-pc:~$python3
Python 3.6.7 (default, Oct 22 2018, 11:32:17)
[GCC 8.2.0] on linux
Type "help", "copyright", "credits" or "license" for more
information.
>>> import tensorflow as tf
>>> print(tf.__version__)
1.14.0
>>> tf.add(2,3)
<tf.Tensor 'Add:0' shape=() dtype=int32>
```

TensorFlow 2.0, on the other hand, evaluates the result that we call the API: